

Phase 7 - README

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PHASE 7 — Risk, Scenario & Monte Carlo Intelligence

Real Estate AI Decision Intelligence Platform

Follows Phase 6 in the Master A-Z Prompt’s 10-phase architecture. Built directly on top of Phase 6’s deterministic financial engine — **reused unchanged, never duplicated** (Master Prompt §7.2). Matches the established code-structure convention: live PostgreSQL via `db/connection.py`, `src/<topic>/*.py` modules, three orchestrators, an automated test suite.

What was built

Module	Purpose	Master Prompt Section
<code>monte_carlo.py</code>	Triangular-distribution Monte Carlo simulation (price, vacancy, opex, appreciation, interest rate), with a documented vacancy↔opex correlation. Every draw runs through Phase 6’s <code>engine.run_financial_engine()</code> unchanged.	7.3, 7.4
<code>risk_metrics.py</code>	IRR/NPV/ROI distributions, probability of negative NPV/IRR, probability of meeting target IRR, VaR/CVaR at 95%/90%, downside percentiles, risk-driver ranking.	7.5
<code>decision_stability.py</code>	Financial-only INVEST/HOLD/AVOID classifier applied per simulated draw; measures how often the Base-case label survives simulation. Not Phase 9’s authoritative decision policy — see docstring.	7.8
<code>scenario_engine.py</code>	Adds Severe Downside to Phase 6’s Base/Upside/Downside.	7.6
<code>batch_runner_risk.py</code>	Runs the full pipeline across a demo sample (reuses Phase 6’s property selection).	—
<code>export_results_csv.py</code>	6 numbered CSVs → <code>docs/risk_results_csv/</code> .	7.9
<code>generate_charts.py</code>	4 charts → <code>docs/risk_charts/</code> .	—
<code>test_phase7.py</code>	21 automated tests (all passing, includes 1 post-delivery audit-fix	7.10

	regression test).	
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Full detail: `docs/phase7_report.md`, `docs/limitations.md`.

Notable finding

Across the entire n=25 demo batch, the **appreciation-rate assumption dominates simulated IRR risk** (mean correlation +0.999) far more than vacancy, opex, or purchase price. This is a structural consequence of Phase 6's low observed rental yields combined with a 5-year holding period — most of the return comes from the exit sale. It also means the Phase 6/Phase 3-documented unreliable-locality-appreciation issue (~14% of localities) matters more for this portfolio's risk than it might otherwise appear. Details in `docs/phase7_report.md` Section 6.

Calibration status

Every result in this phase is UNCALIBRATED. No historical panel data exists in this dataset to fit distribution parameters or correlations against. Distribution bounds are documented judgement calls (see inline comments in `monte_carlo.py`), not statistically derived. Do not present these probabilities as real-world probabilities — see `docs/limitations.md` §1.

How to run

With a live database (production path)

```
cd phase7
pip install -r requirements-phase7.txt --break-system-packages
export DB_USER=...
export DB_PASSWORD=...

python3 src/risk/export_results_csv.py      # docs/risk_results_csv/*.csv
python3 src/risk/generate_charts.py        # docs/risk_charts/*.png
```

Local re-check (no database — what was used to produce this deliverable)

```
cd phase7
pip install -r requirements-phase7.txt --break-system-packages

python3 src/risk/export_results_csv.py --source csv
python3 src/risk/generate_charts.py --source csv

python3 -m pytest src/risk/test_phase7.py -q
```

Single-property example

```
from src.finance.data_loader import load_property_financials_from_csv
from src.finance.engine import FinancialAssumptions
from src.risk.monte_carlo import run_monte_carlo, MonteCarloConfig
from src.risk.risk_metrics import compute_risk_metrics
from src.risk.decision_stability import compute_decision_stability
from src.risk.scenario_engine import run_extended_scenarios, extended_scenario_summary

observed = load_property_financials_from_csv(1)
assumptions = FinancialAssumptions(
    property_id=1,
    annual_operating_expenses=180000,
    vacancy_rate=0.05,
    appreciation_rate=0.08,
)
```

```
mc = run_monte_carlo(assumptions, observed, MonteCarloConfig(n_simulations=2000, seed=42))
risk = compute_risk_metrics(mc)
stability = compute_decision_stability(mc)
scenarios = extended_scenario_summary(run_extended_scenarios(assumptions, observed))
```